

# Linux Directory Structure

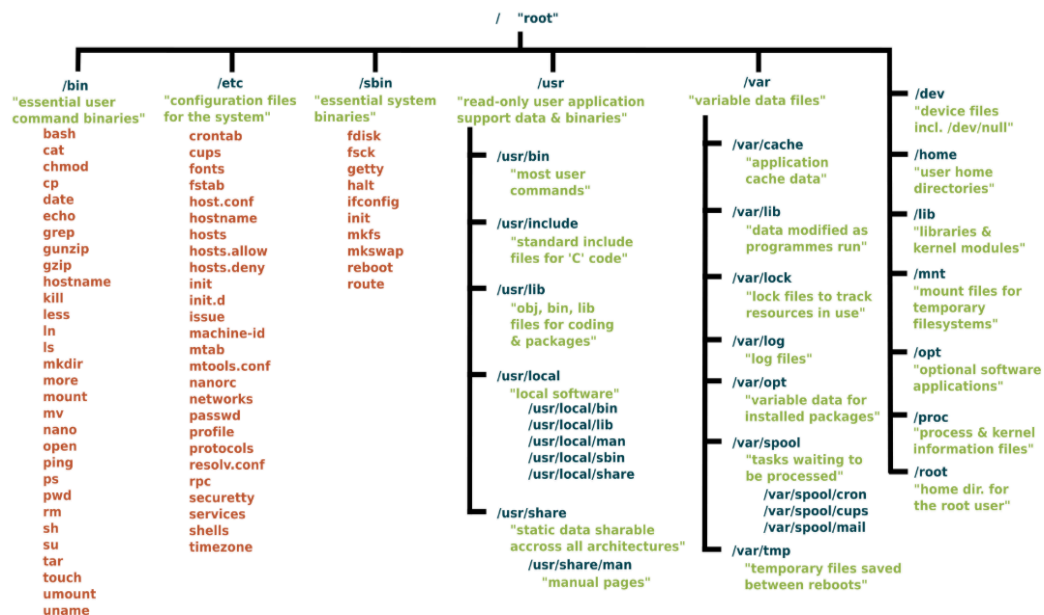
In Linux, the directory structure follows a standard layout called the **Filesystem Hierarchy Standard (FHS)**.

👉 Everything in Linux starts from the **root directory /**

All files, folders, devices, and partitions are organized under this single tree structure.

Think of it like an inverted tree:

```
 /
  |\
   |\
    / | \
   /bin /etc /home ...
```



## 1 / – Root Directory

- The top-most directory in Linux.

- All other directories branch from here.
- Represented by a single forward slash `/`.

⚠ Important:

- Do NOT confuse `/` (root directory) with `/root` (root user's home directory).
  - Only the **root user** has full administrative control over the system.
- 

## **2** `/bin` – Essential User Binaries

Contains basic command binaries required for system operation.

These commands are available to all users and required even in single-user mode.

### Examples:

- `ls`
- `cp`
- `mv`
- `cat`
- `rm`
- `bash`

You can check:

```
ls /bin
```

📌 Note: In modern systems, `/bin` may be linked to `/usr/bin`.

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## **3 /sbin – System Binaries**

Contains system administration commands.

Primarily used by the root user.

### **Examples:**

- `reboot`
- `shutdown`
- `ifconfig`
- `iptables`
- `fsck`

Check:

```
ls /sbin
```

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## **4 /etc – Configuration Files**

Contains system-wide configuration files.

These are mostly text-based configuration files.

### **Important Files:**

- `/etc/passwd` → User account information
- `/etc/shadow` → Encrypted passwords

- `/etc/hosts` → Hostname-to-IP mapping
- `/etc/fstab` → Filesystem mount configuration

Example:

```
cat /etc/hosts
```

📌 Almost every service has its configuration file inside `/etc`.

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## **5** `/home` – User Home Directories

Contains personal directories for normal users.

Example:

```
/home/ashvi  
/home/john
```

Each user has:

- Documents
- Downloads
- Desktop
- User-specific configuration files

When a user logs in, they are automatically placed inside their home directory.

Check:

```
cd ~  
pwd
```

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## **6** `/root` – Root User Home Directory

Home directory of the root (administrator) user.

 Important:

- `/root` is NOT the same as `/`
  - `/` = Root directory
  - `/root` = Root user's personal home directory
- 

## **7** `/var` – Variable Data

Contains files that change frequently.

Used for logs, caches, mail, spool files.

### Important Subdirectories:

- `/var/log` → System logs
- `/var/spool` → Mail, cron jobs
- `/var/cache` → Application cache
- `/var/lib` → Application state data

Example:

```
cd /var/log
ls
```

---

## **8** /tmp – Temporary Files

Stores temporary files.

- Accessible by all users.
- Usually cleared after reboot (depends on system configuration).

Example:

```
cd /tmp
touch testfile
```

Permissions:

```
drwxrwxrwt
```

The **t** means sticky bit (users cannot delete others' files).

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## **9** /usr – User System Resources

Contains user-level applications and utilities.

This is one of the largest directories.

### **Important Subdirectories:**

- **/usr/bin** → User commands (vim, python, gcc)
- **/usr/sbin** → Admin commands
- **/usr/lib** → Libraries
- **/usr/share** → Shared data

Example:

```
which vim
```

Output might show:

```
/usr/bin/vim
```

📌 `/usr` does NOT mean "user home". It means "Unix System Resources".

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## `/lib` and `/lib64` – Libraries

Contain shared libraries required by:

- `/bin`
- `/sbin`

Also contains kernel modules.

Example:

```
ls /lib
```

64-bit systems use:

```
/lib64
```

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## `/boot` – Boot Loader Files

Contains files required to boot the system.

**Important Files:**

- `vmlinux` → Linux kernel
- `initramfs` → Initial RAM filesystem
- `grub/` → Bootloader configuration

Without `/boot`, system cannot start.

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## `/dev` – Device Files

Linux treats hardware as files.

Each hardware device is represented inside `/dev`.

### Examples:

- `/dev/sda` → Hard disk
- `/dev/null` → Discards output
- `/dev/tty` → Terminal

Example:

```
echo "Hello" > /dev/null
```

(Output is discarded.)

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## `/proc` – Process Information (Virtual Filesystem)

Virtual filesystem.



Does NOT store real files on disk.

Contains real-time system and process information.

### Examples:

- `/proc/cpuinfo`
- `/proc/meminfo`
- `/proc/<PID>/`

Example:

```
cat /proc/cpuinfo
```

Shows CPU details.

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## **/sys – System Information**

Another virtual filesystem.

Provides interface to:

- Kernel
- Hardware
- Device drivers

To expose **kernel and hardware information** to users and programs in a structured way.

Used for advanced system and driver management.

### Why Do We Need It?

The Linux kernel controls:

- CPU
- Memory
- Disks
- USB devices
- Network cards
- Drivers

But users and programs need a way to:

- See hardware details
- Check driver status
- Change certain kernel settings

`/sys` is that interface.

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## `/mnt` and `/media` – Mount Points

### `/mnt`

Used for temporary manual mounts.

Example:

```
mount /dev/sdb1 /mnt
```

### `/media`

Used for automatically mounted devices like:

- USB drives

- CDs
- External disks

Example:

```
/media/username/USB_DRIVE
```

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## **/opt – Optional Software**

Used for installing third-party software.

Examples:

```
/opt/oracle
```

```
/opt/tomcat
```

```
/opt/google
```

Often used for custom or enterprise applications.

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## **/run – Runtime Data**

Stores runtime system information since boot.

Replaced older `/var/run`.

Contains:

- PID files
- Runtime service data

Cleared after reboot.

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# Summary Table

| Directory | Purpose                   |
|-----------|---------------------------|
| /         | Root of entire filesystem |
| /bin      | Essential user commands   |
| /sbin     | System admin commands     |
| /etc      | Configuration files       |
| /home     | User home directories     |
| /root     | Root user home            |
| /var      | Logs & variable data      |
| /tmp      | Temporary files           |
| /usr      | User applications         |
| /lib      | Shared libraries          |
| /boot     | Boot files                |
| /dev      | Device files              |
| /proc     | Process information       |
| /sys      | Kernel & hardware info    |
| /mnt      | Temporary mount point     |
| /media    | Auto-mounted devices      |
| /opt      | Optional software         |

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